

5.2.12

EE25BTECH11009 - Anshu kumar ram

Question:

Solve the system of equations

$$2x + y - 6 = 0 \quad (0.1)$$

$$4x - 2y + 4 = 0 \quad (0.2)$$

Solution:

The equation of line:

$$\mathbf{n}^\top \mathbf{x} = c \quad (0.3)$$

Line L:

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 6 \quad (0.4)$$

Line K:

$$\begin{pmatrix} 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -4 \quad (0.5)$$

These can be combined and written in matrix form:

$$\begin{pmatrix} 2 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \quad (0.6)$$

The following augmented matrix can be solved by gaussian elimination

$$\left(\begin{array}{cc|c} 2 & 1 & 6 \\ 4 & -2 & -4 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 1 & 6 \\ 0 & -4 & -16 \end{array} \right) \quad (0.7)$$

From the second row:

$$-4y = -16 \implies y = 4 \quad (0.8)$$

Substituting into the first equation:

$$2x + 4 = 6 \implies x = 1 \quad (0.9)$$

Hence, the solution is

$$(x, y) = (1, 4) \quad (0.10)$$

See Figure,

